

## -- SQL TO COPY WORKS THAT HAVE AT LEAST ONE HEP INSTITUTION

```
COPY (  
WITH authorships_hep AS (  
SELECT DISTINCT  
institution_id[23:]::BIGINT AS institution_idx,  
hep_code  
FROM '/home/lc/m/working/WORKING_ARC_PROJECT/admin_orgs.csv'  
WHERE institution_idx IS NOT NULL AND hep_code IS NOT NULL  
)  
  
SELECT DISTINCT work_idx  
FROM '/home/lc/m/openalex_feb26/parquet/authorships/*.parquet' p  
INNER JOIN authorships_hep  
USING (institution_idx)  
) TO '/home/lc/m/working/WORKING_ARC_PROJECT/processed/works_intermediate_hep.parquet';
```

## -- SQL TO COPY AUTHORSHIPS FOR WORKS WITH AT LEAST ONE HEP

```
COPY (  
SELECT *  
FROM '/home/lc/m/working/WORKING_ARC_PROJECT/processed/works_intermediate_hep.parquet'  
INNER JOIN '/home/lc/m/openalex_feb26/parquet/authorships/*.parquet'  
USING (work_idx)  
WHERE institution_idx IS NOT NULL  
) TO '/home/lc/m/working/WORKING_ARC_PROJECT/processed/authorships_hep.parquet';
```

## -- SQL TO COPY WORKS AND TOPICS FOR WORKS WITH AT LEAST ONE HEP

```
COPY (  
WITH  
topics AS  
(SELECT *  
FROM '/home/lc/m/working/WORKING_ARC_PROJECT/processed/works_intermediate_hep.parquet'  
JOIN '/home/lc/m/openalex_feb26/parquet/topics/*.parquet'  
USING (work_idx)  
)
```

```
SELECT t, "type", source_id, authors_count, institutions_distinct_count, cited_by_count  
FROM topics t  
INNER JOIN '/home/lc/m/openalex_feb26/parquet/works/*.parquet'  
USING (work_idx)  
) TO '/home/lc/m/working/WORKING_ARC_PROJECT/processed/works_hep.parquet'
```

## -- SQL TO CONSTRUCT COUNTS OF DOMAINS PER AUTHOR. COULD USE SAME FOR FIELDS, SUBFIELDS OR TOPICS

```

WITH unnested_counts AS (
-- Step 1: Compute frequencies and unnest the map entries
SELECT
  author_name,
  author_idx,
  count(DISTINCT work_idx) AS works_count,
  unnest(map_entries(histogram(field_name))) AS entry
FROM '/home/lc/m/working/WORKING_ARC_PROJECT/processed/works_hep.parquet' w
INNER JOIN '/home/lc/m/working/WORKING_ARC_PROJECT/processed/authorships_hep.parquet' a
USING (work_idx)
GROUP BY ALL
),
calculated_fractions AS (
-- Step 2: Calculate fractions using a window function
SELECT
  author_name,
  author_idx,
  works_count,
  entry.key AS field,
  entry.value AS field_count,
  -- Divides individual count by the total sum of counts for this specific author
  (entry.value::DOUBLE / SUM(entry.value) OVER (PARTITION BY author_idx)) AS field_fraction
FROM unnested_counts
)
-- Step 3: Re-bundle into a structured list sorted by fraction descending
SELECT
  author_name,
  author_idx,
  works_count,
  list({
    'field': field,
    'count': field_count,
    'fraction': round(field_fraction, 4) -- Rounded to 4 decimal places for clean output
  }) ORDER BY field_fraction DESC AS sorted_fields
FROM calculated_fractions
GROUP BY ALL
ORDER BY works_count DESC;

```